Database Systems - Normalization in DBMS

Here are 10 medium-difficulty multiple-choice questions on **Normalization in DBMS**, designed at the "Apply" level of Bloom’s Taxonomy. These questions require students to analyze table structures and apply normalization rules to solve specific data redundancy or dependency issues.

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Normalization in DBMS: Practice Quiz

1. A relation $R(A, B, C, D)$ has the functional dependencies $A \to B$ and $C \to D$. If we decompose R into $R_1(A, B)$ and $R_2(C, D)$, what is the primary reason this decomposition might be considered "lossy"?

A. It fails to preserve the functional dependencies.

B. It is impossible to reconstruct the original relation R using a natural join.

C. It introduces transitive dependencies.

D. It fails to satisfy the First Normal Form.

Answer: B

2. You are given a table `Student_Courses(StudentID, CourseID, InstructorName, OfficeLocation)`. If `InstructorName` determines `OfficeLocation`, which normal form is violated?

A. First Normal Form (1NF)

B. Second Normal Form (2NF)

C. Third Normal Form (3NF)

D. Boyce-Codd Normal Form (BCNF)

Answer: C

3. A relation $R(A, B, C, D)$ has a primary key $\{A, B\}$. The functional dependencies are $\{A, B\} \to C$ and $A \to D$. Which Normal Form is this relation currently in?

A. 1NF

B. 2NF

C. 3NF

D. BCNF

Answer: A (It violates 2NF because of the partial dependency $A \to D$)

4. When normalizing a database, if you encounter a multi-valued dependency $A \twoheadrightarrow B$, which Normal Form is necessary to resolve it?

A. 3NF

B. BCNF

C. 4NF

D. 5NF

Answer: C

5. Consider a relation `Sales(OrderID, ProductID, Quantity, Discount)`. If `Discount` is determined solely by the `ProductID`, but the primary key is `{OrderID, ProductID}`, how do you transform this into 2NF?

A. Create a new table `Products(ProductID, Discount)` and remove `Discount` from `Sales`.

B. Add `OrderID` to the `Products` table.

C. Move `Quantity` to a separate table.

D. No transformation is needed; it is already in 2NF.

Answer: A

6. If a relation has a candidate key that is a subset of another candidate key, and there is a functional dependency where a non-prime attribute determines part of a candidate key, which normal form is most likely violated?

A. 1NF

B. 2NF

C. 3NF

D. BCNF

Answer: D

7. You have a table with a composite primary key. You observe that an attribute depends on only half of the primary key. Applying the rules of normalization to reach 2NF will result in:

A. Deleting the attribute entirely.

B. Creating a new relation to house the partial dependency.

C. Converting the partial dependency into a multi-valued dependency.

D. Promoting the non-key attribute to a primary key.

Answer: B

8. Given $R(A, B, C)$ with functional dependencies $A \to B$ and $B \to C$. To bring this into 3NF, what is the resulting schema?

A. $R_1(A, B)$ and $R_2(B, C)$

B. $R_1(A, B, C)$

C. $R_1(A, B)$ and $R_2(A, C)$

D. $R_1(A, C)$ and $R_2(B, C)$

Answer: A

9. A database designer is working on a schema where all attributes are atomic, all partial dependencies are removed, and all transitive dependencies are removed. However, there are still anomalies present due to a determinant that is not a candidate key. Which step should the designer take next?

A. Move to 1NF.

B. Move to 2NF.

C. Move to BCNF.

D. Move to 4NF.

Answer: C

10. Why is the decomposition of a relation into BCNF sometimes avoided in practical database design?

A. Because BCNF always leads to data loss.

B. Because BCNF may result in the loss of functional dependency preservation.

C. Because BCNF increases the number of required JOIN operations for every query.

D. Because BCNF is less restrictive than 3NF.

Answer: B